Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/07/2023
PI Name:	Zupanska, Agata Ph.D.		
Project Title:	From Antarctica to Space: Molecular Response and Physiological Adaptation of Moss to Simulated Deep Space Cosmic Ionizing Radiation and Spaceflight Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	zupanska@live.com	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	352-682-5615
Organization Name:	The SETI Institute		
PI Address 1:	189 Bernardo Ave, 200		
PI Address 2:			
PI Web Page:			
City:	Mountain View	**State:**	CA
Zip Code:	94043-5138	**Congressional District:**	18
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	10/22/2021	**End Date:**	10/21/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
Contact Email:	Ye.Zhang-1@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	As budgeted in the original proposal we hired an OPS Technician, Emily Lockwood. Emily is affiliated with University of Florida and paid from the subaward. Emily works on the project accordingly to proposed effort, 6 months for each year of grant project.		
COI Name (Institution):	McDaniel, Stuart Ph.D. (University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC22K0208		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Rationale: Deep space environments of cosmic ionizing radiation (cIR) from astrophysical events and microgravity (micro-g) are outside of plant evolutionary experience and can impede the vital role of plants in space exploration. Understanding the combined effects of cIR and micro-g on plants at the molecular level is pivotal to engineering deep space-tolerant plants for human space exploration. Most previous research studies each condition alone and uses seeds irradiated with high doses of acute cIR. Hypothesis and Objectives: We propose a Research Investigation (Project Type) into plant growth, development, and physiology (Focus Area) using an International Space Station (ISS)-Flight+Ground Study Component (Platform) to test the hypothesis that combined deep space cIR and micro-g affect plants differently and more profoundly than each environment alone, resulting in distinct molecular responses and physiological adaptation. We aim to detect how the effects of combined simulated chronic deep space cIR and sequential spaceflight micro-g differ from those of cIR or micro-g alone on live plant growth, development, and physiology, as reflected by changes in gene expression, to identify molecular response and physiological adaptation to these environments. Research Approaches and Specific Aims: We will use Antarctic moss Ceratodon purpureus (C. purpureus) which allows us to use irradiated live plants rather than spores in spaceflight. Ceratodon purpureus is tolerant of extreme conditions and is ideal for human spaceflight missions due to its small size, lightweight spores, low-maintenance, and roles in oxygenation, regolith enrichment, and medicine production. We will use a twofold approach: 1) in the ground study protonema will be exposed to simulated Galactic Cosmic Rays (GCRSim, 10 and 100 cGy) and Solar Particle Events (SPESim, 100-1000 cGy) delivered at fractionated dose; 2) in the ISS-Flight study protonema exposed to GCRSim or SPESim on the ground and housed in the NASA Biological Research in Canisters (BRIC)-Light Emitting Diode (LED) will be launched to the ISS and grown in low-Earth orbit (LEO). We will use morphological metrics, RNAseq, and bioinformatics to achieve: AIM1- Discover the dose- and time-dependent effects of simulated deep space cIR on growth, development, and gene expression in C. purpureus. AIM2- Identify the different effects on growth, development, and gene expression in C. purpureus of simulated deep space cIR – singular, or combined with spaceflight micro-g. AIM3- Identify the different effects on growth, development, and gene expression in C. purpureus of spaceflight micro-g – singular, or combined with simulated deep space cIR. Outcome and Significance: This study will: 1) analyze moss growth and development in cIR and micro-g, which will inform plant performance capabilities for human spaceflight exploration; 2) discover moss response and physiological adaptation to cIR and micro-g at the gene expression level, and by extension in other biological systems; 3) discover candidate genes for genetic manipulation to engineer deep-space tolerant plants which will advance the design of Bioregenerative Life Support systems and plant habitats for planetary missions; and 4) extend our results to other multicellular organisms, which will inform health risk management. Relevance to the Plant Biology Element of the NASA Space Biology program: This project addresses: a) the main goal, to “understand plant growth in spaceflight environments and physiological responses to those environments;” b) guiding questions #1 - How does gravity affect plant growth, development and metabolism; #4 - What are the effects of chronic exposure to cosmic radiation on plants; and #5 How do plants sense and react to gravity and what are the molecular mechanisms involved; and c) Plans and Research Solicitations objectives #1 Determine how plants adapt to spaceflight from the molecular to whole plant level using existing hardware (BRIC) on ISS; #3 - Answer basic research questions about plant biology that will enable the future development of a bioregenerative life support system for use by humans in space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In our study we deploy novel approaches in plant space biology research enabling to maximize scientific return. First, unlike other studies relying on model plants, we bring a new organism, moss, which is remarkably adaptive to extreme environments here on Earth and thus is likely to exhibit superior adaptability to deep space environments. Second, in contrast to most other studies which address each deep space environment separately, we will discover the combined effects of both cosmic ionizing radiation and spaceflight microgravity. Third, in contrast with previous research done on irradiated seeds and spores which leads to mutagenesis rather than observation of the effects of deep space cosmic ionizing radiation on plant physiology, our pioneering experimental design sequentially exposes live, actively growing plants to deep space environments. Fourth, we deploy the most accurate simulation of cosmic ionizing radiation of astrophysical origin, exposure to ion beams of compositions and energies best approximating Galactic Cosmic Rays (GCR) and Solar Particle Events (SPE) radiation. Most studies used exposure to electromagnetic wave radiation from UVB, UVC, X-rays, and Gamma-rays delivered at a high-dose in a single pulse (acutely), which poorly simulates deep space cosmic ionizing radiation, which is particulate, chronic, and low-dose at extended fractionations. Upon completion we will deliver moss transcriptomic profiles, growth and development analysis and candidate genes for genetic manipulation to engineer plants more tolerant of cosmic ionizing radiation and spaceflight microgravity. Firstly, this will have a direct impact on advancing plant preparedness for space exploration missions and will help constructing plant varieties that can be incorporated into life support systems and in-space human habitation, enabling long-duration crewed missions beyond Low Earth Orbit. Secondly, since our study will identify the critical plant mechanisms of prevention or repair of the damage caused by ionizing radiation it will potentially result in developing moss-based radiation countermeasures agents of three classes: protectors, mitigators and therapeutics. Such plant compounds could be then used by astronauts during long term interplanetary travel as well as administered to radiation oncology patients or nuclear plant workers accidentally exposed to high radiation doses here on Erath.

Task Progress:

1. Establishing moss ANT1 tolerance of high levels of gamma ray radiation.
To successfully execute the NASA Biological Research in Canisters (BRIC)-27 experiment, our Antarctic moss ANT1 needs to tolerate cosmic-like ionizing radiation delivered in ion beams: simulated Galactic Cosmic Rays (GCRSim) at 20, 80cGy, and Solar Particle Events (SPESim) at 40, 200cGy. Before having access to the accelerator at the NASA Space Radiation Laboratory (NSRL) at the Brookhaven National Laboratory (BNL) (Spring campaign 2023), we used the onsite Cs-137 irradiator (UF) to titrate moss tolerance of high levels of ionizing gamma ray radiation. We exposed the Petri plates with ANT1 spot-inoculum colonies, prepared in the exact same way as will be those used in BRIC-27 to 10, 100, 500, and 1200Gy of gamma rays, at a moderate (0.45Gy/min) dose rate. Post-irradiation, one set of Petri plates was grown in LIGHT and RT, while the second set was kept in DARK and COLD to recreate the light and temperature conditions during a wait for the spaceflight launch, pre-flight ops, launch, orbiting, and berthing. The LIGHT and RT set established the post-exposure growth rate in moss grown in optimal conditions, while the DARK and COLD set tested if moss exposed to ionizing radiation and prevented from immediate damage repair due to cold stowage, as planned for BRIC-27, would be negatively impacted. Gamma ray radiation at 10, 100, and 500Gy dose had no adverse effects on the ANT1 growth and health in either growth condition. Although the growth rate of ANT1 exposed to 1200Gy was partially inhibited at 5d post-irradiation, at 8d ANT1 showed signs of recovery and increase in growth rate, and at 12d, ANT1 was robustly growing. This indicates that gamma ray radiation at the 1200Gy dose had some adverse effects on ANT1, but this dose was not yet lethal. This is relevant to confirming the appropriateness of the BRIC-27 experimental design and intended stowage of irradiated ANT1 in dark and cold conditions until on the International Space Station (ISS).

2. Exposure of moss ANT1 to SPESim – the piggyback experiment at NSRL, BNL (23A Spring 2023 campaign).

Moss ANT1 on spot-inoculum plates was exposed to SPESim ion beams at the NSRL, BNL. This experiment was conducted as a piggyback experiment in the 23A campaign (April 19, 2023) courtesy of Dr. Ye Zhang, NASA Kennedy Space Center (KSC). Moss on the Petri plates was exposed to SPESim at 40, 80, and 200cGy at the low dose rate (not exceeding 9cGy/min). Post-exposure, the Petri plates were kept in DARK and COLD for 3 weeks and then allowed to grow for 7d in LIGHT and RT. This recreated the exact timeline intended for the BRIC-27 experiment. ANT1 showed no signs of being exposed to SPESim protons at either dose. Also, ANT1 was not affected by prolonged cold stowage following exposure to ion beams. When returned to the LIGHT and RT, ANT1 was robustly growing, and all the colonies were green and fuzzy. This indicates that: 1.) ANT1 tolerates the SPESim doses intended to be used in the BRIC-27 experiment (40 and 200cGy); 2.) ANT1 stowage in dark and cold immediately post-exposure, although slows down the metabolism, does not compromise moss viability after receiving high levels of particulate radiation; 3.) ANT1 transferred to the optimal growth conditions (LIGHT and RT), recovers from exposure to the particulate radiation and grows robustly.

Based on the outcome from the piggyback experiment, we concluded that the original experimental design of BRIC-27 was suitable and that the chance for successful science return from the BRIC-experiment was very high.

3. The BRIC-27 SVT.

3.1. The SVT overview. The main objectives of the Science Verification Test (SVT) were to assess moss viability, morphology, and level of contamination, as well as the quantity and quality of total RNA, while best replicating moss growth timeline and exposure to simulated cosmic ionizing radiation scheduled for the BRIC-27 experiment to the ISS. The SVT was entirely performed at the Co-Investigator (Co-I) lab (University of Florida/UF) with the following modifications:

1.) GCRSim and SPESim exposures at BNL were replaced with exposure to gamma rays from Co-60 and Cs-137 sources using irradiators located at the UF campus. The appropriate quality factors for each type of particulate radiation were used to calculate corresponding doses of gamma ray electromagnetic radiation, a quality factor of 4.0 for GCRSim and 1.5 for SPESim. Therefore: - 80cGy of gamma rays from Co-60 recreated GCRSim at 20cGy - 320cGy of gamma rays from Cs-137 recreated GCRSim at 80cGy - 60cGy of gamma rays from Co-60 recreated SPESim at 40cGy - 300cGy of gamma rays from Cs-137 recreated SPESim at 200cGy

2.) Confinement to BRIC/PDFU hardware was replaced by vacuum sealing of individual Petri plates in food storage bags. A set of Electric Vacuum Sealer and Reusable Vacuum Food Storage Bags (KOSBON) was used. The air surrounding a Petri plate was removed, leaving a thin layer of air trapped inside a Petri plate. This replicated limited gas exchange as in air-tight locked Petri Dish Fixation Unit (PDFU).

3.2. The SVT execution. The Petri plates with ANT1 spot-inoculum colonies were subjected to the entire timeline as designed for the BRIC-17 experiment: 1. Moss spot-inoculum Petri plate preparation 2. 7-day growth in LIGHT at 25°C 3. 7-day growth in DARK at RT (23°C) – recreating transport to NSRL, BNL, pre-irradiation ops 4. Exposure to gamma rays – recreating exposures to GCRSim and SPESim 5. 3-week storage in DARK and COLD (4°C) – recreating a wait for the spaceflight 6. Vacuum sealing – recreating integration into payload hardware (BRIC/PDFU) 7. 7-day storage in DARK and COLD (4°C) – recreating pre-flight ops, launch, orbiting, berthing with the ISS 8. 7-day growth in DARK and RT (23°C) – recreating growth on the ISS 9. Dispensing RNAlater and resealing in vacuum bags – recreating in-flight chemical fixation 10. 24hr fixation at RT (23°C) – recreating fixation in RNAlater in orbit 11. 7-day storage in -80°C freezer – recreating storage in Minus Eighty-Degree Laboratory Freezer (MELFI) on the ISS 12. Sample analysis – recreating post-flight sample analysis 13. Report (7d+7d+21d+7d+7d+1d+7d=57d, +5 calendar days for RNA analysis and writing) delivered within 62d after initiating the SVT.

3.3. The SVT science performance. The findings of the BRIC-27 SVT were evaluated against the success criteria as formulated in the BRIC-27 Experimental Research Design (ERD) document.

1. PLANT MOSS/VIABILITY - EXCELLENT 100% (39) moss spot-inoculum per treatment, non-irradiated and irradiated, with all 13 spot-inoculum colonies per plate remained viable up to fixation in RNAlater. 2. PLANT MOSS/MORPHOLOGY - EXCELLENT 100% (39) of moss spot-inoculum colonies per treatment, non-irradiated and irradiated, with all 13 spot-inoculum colonies per plate showed green healthy appearance, no signs of necrotic lesions, wilting, or other morphological aberrations. 3. BACTERIAL OR FUNGAL CONTAMINATION - EXCELLENT No bacterial or fungal contamination on any plate; 0% of moss spot-inoculum colonies showed contamination with bacteria or fungi per treatment. 4. QUANTITY OF TOTAL RNA EXTRACTED - EXCELLENT Above 400ng of total RNA extracted from 4 spot-inoculum colonies for each replicate. 5. QUALITY OF TOTAL RNA EXTRACTED - ACCEPTABLE/UNACCEPTABLE - Three treatments – mild to moderate RNA degradation, RNA with RNA Integrity Number (RIN) between 4-7 for 2 replicates and 1 replicate RIN<4; - One treatment – mild to moderate RNA

degradation, RNA with RIN between 4-7 for 1 replicate and 2 replicates RIN<4; - One treatment – severe RNA degradation, RNA with RIN below ?4 extracted for each replicate.

4. Mitigation Strategies

After completion of the BRIC-27 SVT and identifying the problems with ANT1 moss tissue fixation in RNAlater when “integrated into PDFUs”, we focused our efforts on achieving the efficient ANT1 moss tissue fixation suitable for the spaceflight experiment.

4.1. Improving ANT1 moss fixation in RNAlater.

4.1.1. Identifying the cause of poor ANT1 moss fixation in RNAlater. Upon submersion of ANT1 moss tissue in RNAlater, we observed air bubble retention on moss filaments indicative of super-hydrophobic tissue surface. This is unexpected of plants originated in Antarctica, which tend to have high wax content within the cuticle. The behavior of polar liquids, such as RNAlater, in contact with plant hydrophobic surfaces impedes: 1.) liquid adherence to the surface; 2.) liquid absorption to the surface; 3.) liquid permeation into the tissue.

4.1.2. Developing strategies to increase RNAlater adherence and absorption on ANT1 hydrophobic surfaces. Surfactants reduce air bubble surface tension. Addition of surfactants to RNAlater was expected to destabilize and dislodge air bubbles, thus improve RNAlater adherence and absorption. We tested four different surfactants: non-ionic Tween-20 and NP-40S, cationic CTAB, and anionic Alcanox. The addition of small volumes of either surfactant to RNAlater had noticeable effects on air bubble behavior and caused air bubbles to round up, shrink, and/or detach. However, total RNA extracted from the moss tissue fixed in RNAlater with addition of either surfactant, was not of good quality; neither was the quantity satisfactory. This indicated that improving RNAlater adherence and improving RNAlater absorption to the moss tissue was not sufficient to achieve effective fixation in RNAlater and pointed toward inefficiency in RNAlater permeation through the moss tissue.

4.1.3. Developing strategies to increase RNAlater permeation into the ANT1 moss tissue. Oil-based adjuvants act on lipid solubilization and relaxing of a plant cuticle structure (softening), thus aiding penetration of low affinity polar solvents. RNAlater in the presence of the petroleum oil-based adjuvant preserved moss RNA considerably better than when mixed with orange oil-based adjuvant, suggesting that paraffin oil, but not orange oil, improved RNAlater permeation through the ANT1 cuticular waxes. The efficiency of ANT1 fixation in RNAlater mixed with petroleum oil-based adjuvant was further increased by shortening of the fixation time from 24h to 30min at RT (ISS ambient).

4.2. Identifying 70% Ethanol as an alternative chemical fixative to RNAlater.

Ethanol increases permeation on hydrophobic and superhydrophobic surfaces. Intracellularly, ethanol causes the water molecules in the tissue to be bound by its hydroxyl groups, so the moisture in the tissue is slowly removed and replaced by ethanol. Enzyme dehydration by ethanol is thus a viable alternative to protein precipitation by high salts as occurring during fixation in RNAlater. ANT1 tissue submersed in ethanol between 30-99.9% concentrations never retained air bubbles, suggesting that ethanol adherence and absorption to tissue was not impacted by the ANT1 hydrophobic surfaces. Total RNA obtained from moss tissue fixed in 70% was of a very high quality, from that fixed in 50% ethanol of moderate quality, while in 30% ethanol was fully degraded. The large quantity of total RNA extracted from moss tissue fixed in 70% ethanol further supported the efficient fixation at 70% concentration. Ethanol, although proven very successful in chemical fixation of ANT1 moss tissue, is also flammable, with a relatively low flash point temperature, and thus has not been approved by the NASA Project Team as a suitable fixative to be used in BRIC-27.

4.3. Modifying the experimental design: substituting BRIC-27 with ARTEMOSS. Since ANT1 moss tissue fixation in RNAlater was proven difficult and ethanol is not the most suitable solvent to use in spaceflight, we decided to eliminate the chemical fixation step from the BRIC-27 experiment and replace it with the fast deep-freezing of moss tissue on Petri plates. A fast-deep-freezing method is the best method to preserve plant tissue; however, many times the access to liquid nitrogen or deep freezer is limited or impractical. We confirmed that the fast deep-freeze can be achieved on the ISS by deploying ice blocks prechilled in the GLACIER (-160°C). We have exceptionally good results with extracting total RNA from frozen, never thawed ANT1 tissue and we consistently extract RNA of the highest quality at the largest quantities, rivaling only total RNA obtained from fresh, never frozen tissue. Additionally, we tested and confirmed outstanding RNA purity and stability obtained from frozen non-fixed ANT1 tissue. The recommendations have been made to the NASA project team and we are awaiting the evaluation:

1. Eliminate BRIC/PDFU hardware: moss tissue on Petri plates not housed in hardware will freeze considerably faster; 2. Build a custom-made Petri plate holder to secure Petri plates in fixed orientation and configuration; 3D-printing of Petri plate rack is encouraged; 3. Use of deep-frozen freeze blocks, ice packs to speed moss tissue freezing; 4. Maintain moss tissue in the frozen state; since moss tissue would not be chemically fixed, the tissue CANNOT THAW until processed at the PI laboratory; 5. Modifications to the biology “integration into hardware section" of the BRIC-27 design; 6. Modifications to the BRIC-27 experiment timeline on the “operation in orbit section”; all other experimental steps will not differ from the original design; 7. Change of the experiment name from BRIC-27 to ARTEMOSS ("ANT1 Radiation Tolerance Experiment with Moss in Orbit on the Space Station").

Bibliography Type:	Description: (Last Updated: 07/17/2025)
Abstracts for Journals and Proceedings	Zupanska AK, Lockwood E, Zhang Y, McDaniel SF. "Moss tolerance of deep space-like ionizing radiation, singular or combined with spaceflight microgravity- preparation for the BRIC-27 experiment to the ISS." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Articles in Peer-reviewed Journals	Zupanska AK, Arena C, Zuñiga GE, Casanova Katny MA, Turnbull JD, Bravo LA, Ramos P, Sun H, Shishov VV. "Editorial: Revisiting the limits of plant life - plant adaptations to extreme terrestrial environments relating to astrobiology and space biology." Front. Plant Sci. 2023 Aug 17;14:1267183. https://doi.org/10.3389/fpls.2023.1267183 , Aug-2023